Supplemental Table ST1. Genomic DNA input concentrations, total DNA input, and PCR amplification cycles used for Oxford Nanopore library preparation. The table summarizes the amount of genomic DNA available from each sample prior to library preparation, the DNA input used for sequencing library construction, and the number of PCR amplification cycles applied. These data are provided to document sample-to-sample variation in starting DNA quantity and amplification requirements and to facilitate interpretation of differences in sequencing read recovery among samples.

Sample Name	Location name as shown in Table 1	Filter type	Sample Code	Qubit [ng/uL]	Lib [ng/uL]	[ng] Input into Pool	Total ng Loaded onto Minion	# of PCR Cycles	Number of Reads 9.4 FlowCell	Average read length (bp)
FS_W_TH_rep1 (1)	Five Sisters Spring 3	Sterivex	JY01	0.63	12.6	15.00	5.55	16	18,359	3,771
FS_W_TH_rep2 (1)	Five Sisters Spring 3	Sterivex	JY02	0.46	12	15.00	5.55	16	9,025	3,919
FS_W_TH_rep3 (1)	Five Sisters Spring 3	Sterivex	JY03	0.6	4.9	15.00	5.55	16	10,292	3,806
FS_W_TM_rep1 (2)	Five Sisters Spring 1	Sterivex	JY04	0.49	8.1	15.00	5.55	16	9,455	3,870
FS_W_TM_rep2 (2)	Five Sisters Spring 1	Sterivex	JY05	0.52	10.5	15.00	5.55	16	8,536	3,736
FS_W_TM_rep3 (2)	Five Sisters Spring 1	Sterivex	JY06	0.56	5.7	15.00	5.55	16	10,246	3,514
FS_W_TL_rep1 (3)	FSS effluent channel	Sterivex	JY07	0.92	7.4	15.00	5.55	16	7,076	3,598
FS_W_TL_rep2 (3)	FSS effluent channel	Sterivex	JY08	0.87	15.2	15.00	5.55	16	5,974	3,285
FS_W_TL_rep3 (3)	FSS effluent channel	Sterivex	JY09	0.95	9.6	15.00	5.55	16	6,482	3,781
OS_W_rep1 (16)	OS effluent channel	Sterivex	JY46	1.51	47.00	10.00	2.00	16	70,791	3,506
OS_W_rep2 (16)	OS effluent channel	Sterivex	JY47	1.3	2.14	10.00	2.00	16	57,158	3,549
OS_W_rep3 (16)	OS effluent channel	Sterivex	JY48	1.26	10.50	10.00	2.00	16	77,300	3,093
OS_W_2nd _rep1 (19)	Octopus Spring	Sterivex	JY55	0.28	35.60	10.00	3.30	18	33,503	3,233
OS_W_2nd_rep2 (19)	Octopus Spring	Sterivex	JY56	0.44	16.90	40.00	10.60	16	21,303	3,978
OS_W_2nd_rep3 (19)	Octopus Spring	Sterivex	JY57	0.38	29.00	40.00	10.60	16	15,567	3,646
FS_CTL_W_rep1 (14)	Filter control	Sterivex	JY40	BDL	BDL	>0.25	>0.06	22	14	3,072
FS_CTL_W_rep2 (14)	Filter control	Sterivex	JY41	BDL	BDL	>0.25	>0.06	22	6	2,945
FS_CTL_W_rep3 (14)	Filter control	Sterivex	JY42	BDL	BDL	>0.25	>0.06	22	24	2,558
FS_W_TH_PC_rep1 (1)	Five Sisters Spring 3	PC	JY61	0.15	3.9	2	0.84	18	10,608	2,333
FS_W_TH_PC_rep2 (1)	Five Sisters Spring 3	PC	JY62	1	42.00	40.00	10.60	16	74,009	3,023
FS_W_TM_PC_rep1 (2)	Five Sisters Spring 1	PC	JY65	0.36	7.60	40.00	10.60	16	51,899	3,086
FS_W_TM_PC _rep2(2)	Five Sisters Spring 1	PC	JY66	0.21	4.88	10.00	3.30	18	52,268	2,940
FS_W_TL_PC_rep1 (3)	FSS effluent channel	PC	JY69	0.22	2.2	10	3.30	18	48,640	2,742
FS_W_TL_PC_rep2 (3)	FSS effluent channel	PC	JY70	0.15	3.86	2	0.84	18	6,861	2,387
OS_W_PC_rep1 (16)	OS effluent channel	PC	JY73	1.58	16.40	40.00	10.60	16	20,505	3,868
OS_W_PC_rep2 (16)	OS effluent channel	PC	JY74	0.87	13.00	40.00	10.60	16	26,530	3,573
OS_W_2nd_PC_rep1 (19)	Octopus Spring	PC	JY75	BDL	BDL	<0.08	<0.05	22	-	-
OS_W_2nd_PC_rep2 (19)	Octopus Spring	PC	JY76	0.75	21.60	40.00	10.60	16	70,722	3,337
FS_CTL_W_PC (14)	FilterControl	PC	JY77	BDL	BDL	<0.08	<0.05	22	15	1,886
FS_W_TH_SF_rep1 (1)	Five Sisters Spring 3	Supor	JY63	0.32	26.60	10.00	3.30	18	72,838	2,868
FS_W_TH_SF_rep2 (1)	Five Sisters Spring 3	Supor	JY64	0.3	11.7	2	0.84	18	15,079	2,371
FS_W_TM_SF_rep1 (2)	Five Sisters Spring 1	Supor	JY67	0.27	15.10	10.00	3.30	18	56,039	2,866
FS_W_TM_SF_rep2 (2)	Five Sisters Spring 1	Supor	JY68	0.31	9.90	10.00	3.30	18	57,869	2,869
FS_W_TL_SF_rep1 (3)	FSS effluent channel	Supor	JY71	1.66	12.00	40.00	10.60	16	35,318	3,590
FS_W_TL_SF_rep2 (3)	FSS effluent channel	Supor	JY72	1.05	32.20	40.00	10.60	16	54,465	2,970
FS_CTL_W_SF (14)	FilterControl	Supor	JY78	BDL	0.14	0.21	0.12	22	979	1,244

Supplemental Table ST2. Genomic DNA input concentrations, total DNA input, and PCR amplification cycles used for Illumina library preparation. The table summarizes the amount of genomic DNA available from each sample prior to library preparation, the DNA input used for sequencing library construction, and the number of PCR amplification cycles applied. These data are provided to document sample-to-sample variation in starting DNA quantity and amplification requirements and to facilitate interpretation of differences in sequencing read recovery among samples.

Sample Name	Location name as shown in Table 1	Filter type	Sample Code	DNA [ng/µL]	PCR Cycles	Library Conc.	0.25ng eqv (5µL max)	15ng eqv	Number of reads
FS_W_TH_rep1 (1)	Five Sisters Spring 3	Sterivex	JY01	0.63	11	11.70		1.28	989,563
FS_W_TH_rep2 (1)	Five Sisters Spring 3	Sterivex	JY02	0.46	11	11.00		1.36	900,236
FS_W_TH_rep3 (1)	Five Sisters Spring 3	Sterivex	JY03	0.6	11	12.50		1.20	1,413,658
FS_W_TM_rep1 (2)	Five Sisters Spring 1	Sterivex	JY04	0.49	11	12.80		1.17	1,682,435
FS_W_TM_rep2 (2)	Five Sisters Spring 1	Sterivex	JY05	0.52	11	11.70		1.28	1,058,431
FS_W_TM_rep3 (2)	Five Sisters Spring 1	Sterivex	JY06	0.56	11	10.60		1.42	1,358,404
FS_W_TL_rep1 (3)	FSS effluent channel	Sterivex	JY07	0.92	11	14.90		1.01	578,635
FS_W_TL_rep2 (3)	FSS effluent channel	Sterivex	JY08	0.87	11	14.90		1.01	445,564
FS_W_TL_rep3 (3)	FSS effluent channel	Sterivex	JY09	0.95	11	8.60		1.74	1,536,703
OS_W_rep1 (16)	OS effluent channel	Sterivex	JY46	1.51	11	14.80		1.01	757,089
OS_W_rep2 (16)	OS effluent channel	Sterivex	JY47	1.3	11	15.10		0.99	666,521
OS_W_rep3 (16)	OS effluent channel	Sterivex	JY48	1.26	11	28.90		0.52	414,310
OS_W_2nd _rep1 (19)	Octopus Spring	Sterivex	JY55	0.28	11	12.70		1.18	1,217,240
OS_W_2nd_rep2 (19)	Octopus Spring	Sterivex	JY56	0.44	11	9.00		1.67	1,602,692
OS_W_2nd_rep3 (19)	Octopus Spring	Sterivex	JY57	0.38	11	12.90		1.16	1,469,031
FS_CTL_W_rep1 (14)	Filter control	Sterivex	JY40	BDL	11	0.05	9.40		9,314
FS_CTL_W_rep2 (14)	Filter control	Sterivex	JY41	BDL	11	0.25	2.00		6,920
FS_CTL_W_rep3 (14)	Filter control	Sterivex	JY42	BDL	11	0.07	7.20		82,713
FS_W_TH_PC_rep1 (1)	Five Sisters Spring 3	PC	JY61	0.15	11	4.90		3.06	1,916,288
FS_W_TH_PC_rep2 (1)	Five Sisters Spring 3	PC	JY62	1	11	1.98		7.58	1,374,688
FS_W_TM_PC_rep1 (2)	Five Sisters Spring 1	PC	JY65	0.36	11	7.60		1.97	2,504,075
FS_W_TM_PC _rep2(2)	Five Sisters Spring 1	PC	JY66	0.21	11	8.10		1.85	2,054,662
FS_W_TL_PC_rep1 (3)	FSS effluent channel	PC	JY69	0.22	11	6.30		2.38	1,632,960
FS_W_TL_PC_rep2 (3)	FSS effluent channel	PC	JY70	0.15	11	3.60		4.17	948,092
OS_W_PC_rep1 (16)	OS effluent channel	PC	JY73	1.58	11	4.70		3.19	825,527
OS_W_PC_rep2 (16)	OS effluent channel	PC	JY74	0.87	11	1.90		7.89	745,045
OS_W_2nd_PC_rep1 (19)	Octopus Spring	PC	JY75	BDL	11	0.03	5.00		-
OS_W_2nd_PC_rep2 (19)	Octopus Spring	PC	JY76	0.75	11	4.10		3.66	1,859,461
FS_CTL_W_PC (14)	FilterControl	PC	JY77	BDL	11	0.04	5.00		16,298
FS_W_TH_SF_rep1 (1)	Five Sisters Spring 3	Supor	JY63	0.32	11	6.70		2.24	1,906,507
FS_W_TH_SF_rep2 (1)	Five Sisters Spring 3	Supor	JY64	0.3	11	7.60		1.97	2,264,944
FS_W_TM_SF_rep1 (2)	Five Sisters Spring 1	Supor	JY67	0.27	11	7.30		2.05	2,044,358
FS_W_TM_SF_rep2 (2)	Five Sisters Spring 1	Supor	JY68	0.31	11	5.90		2.54	1,893,121
FS_W_TL_SF_rep1 (3)	FSS effluent channel	Supor	JY71	1.66	11	7.30		2.05	1,055,202
FS_W_TL_SF_rep2 (3)	FSS effluent channel	Supor	JY72	1.05	11	8.30		1.81	944,559
FS_CTL_W_SF (14)	FilterControl	Supor	JY78	BDL	11	0.03	5.00		9,768

BDL: Below Detection Limit

High pool was diluted from 7.9 ng/µL to 1.8 ng/µL

Low pool was 0.13 ng/µL

Pools were combined at a 1:9 ratio (low:high), with a final pooled concentration of 0.78 ng/µL to be sequenced.

Supplemental File ST3. ONT-based shotgun metagenomic sequencing reads assigned to microbial taxa in YNP samples collected using different filtration methods.

Species	Domain	YNP01	YNP02	YNP03	YNP04	YNP05	YNP06	YNP07	YNP08	YNP09	YNP46	YNP47	YNP48	YNP55	YNP56	YNP57	YNP40	YNP41	YNP42
		FSS3 (Sterivex)			FSS1 (Sterivex)			FSSe2 (Sterivex)			OSe (Sterivex)			OS (Sterivex)			Sterivex control		
Pyrobaculum sp. WP30	Archaea	211	125	137	70	81	81	185	143	130	50	52	18	723	543	851			
Pyrobaculum sp. OCT_11	Archaea	10	12	17	2	7	8	18	17	13	5	7	1	55	52	82			
Thermoproteus sp. CP80	Archaea	33	12	13	14	7	13	25	27	30	6	6	1	43	31	62			
Pyrobaculum sp. JCHS_4	Archaea	15	7	12	8	4	2	12	11	5	4	3		42	27	33			
Pyrobaculum ferrireducens	Archaea	19	4	6	5	2	7	9	8	5	2	2		39	24	36			
Thermoprotei archaeon	Archaea	18	13	16	9	14	12	15	16	16	7	8	2	10	18	17			
Desulfurococcaceae archaeon AG1	Archaea	33	3	11	8	11	4	22	16	12	4	8		9	8	7			
Pyrobaculum aerophilum	Archaea		1			1	2	3	2	2	1	1	1	3	5	9			
Thermoproteus sp. JCHS_4	Archaea	2	1	2		2	1	2		3	1			9	8	7			
archaeon HR02	Archaea	6		6	3	6	2	10	6	11	12	13	4	9	9	16			
Caldivirga sp. MU80	Archaea	4	1	4	1	6	3	6	4	4				7	7	8			
Candidatus Caldiarchaeum subterraneum	Archaea	2				1		10	9	16	89	151	60	2	2	4			
Pyrobaculum islandicum	Archaea		3	2	2	3	3	2		1	1			3	3	7			
Thermocrinis ruber	Bacteria	17412	8538	9696	8923	8028	9651	5090	4288	4855	4108	4923	2466	67832	54675	73802			
Synechococcus sp. JA-2-3B'a(2-13)	Bacteria							506	468	368	19023	9508	7973	9	4	8			
Thermus aquaticus	Bacteria			3	4	3	2	21	17	13	289	84	178	12	19	17			
bacterium HR17	Bacteria	54	30	31	42	41	53	47	30	34	111	111	58	422	372	480		1	
Clostridioides difficile	Bacteria												1						
Candidatus Thermochlorobacter aerophilum	Bacteria						1	139	122	134	393	315	192		2				
Thermocrinis sp. GBS	Bacteria	173	95	121	99	94	124	48	37	53	43	49	24	682	519	751			
Armatimonadetes bacterium CP1_7O	Bacteria			1	1		1	8	9	8	307	98	143	2	1	4			
bacterium HR21	Bacteria			4	2			42	26	29	813	454	521	6	7	6			
Armatimonadetes bacterium DC	Bacteria		1					11	6	8	207	84	105	1	2	2			
Synechococcus sp. JA-3-3Ab	Bacteria							36	40	28	1269	637	521						
Bacteroidetes bacterium	Bacteria			2	6	2	4	177	146	133	500	408	228	3	1	3			
Thermomicrobium roseum	Bacteria							13	10	16	383	246	283	2	2	3			
Armatimonadetes bacterium	Bacteria				1			4	8	8	136	40	79			3			
Roseiflexus sp. RS-1	Bacteria							17	13	16	593	275	195	1		1			
bacterium HR16	Bacteria		2	1	2	3		8	5	17	430	375	233	11	6	6			
Synechococcus sp. 65AY6A5	Bacteria							28	25	25	898	448	393						
Thermoflexus hugenholtzii	Bacteria						2	2		2	37	8	10	1		1			
Gloeomargarita lithophora	Bacteria							38	32	34	205	153	87						
Ralstonia solanacearum	Bacteria								1		1	1	2						
Lactobacillus rhamnosus	Bacteria																		
Synechococcus sp. 63AY4M2	Bacteria							28	25	20	888	469	381	1	1				
bacterium HR15	Bacteria							2	3	6	182	120	78	1	2				
Chloracidobacterium thermophilum	Bacteria		1	2		1	1	17	23	10	16	8	10	1	3	3			
Acidobacteria bacterium	Bacteria		1	2	2		3	67	45	65	158	133	79	8	8	19			
Synechococcus sp. 65AY6Li	Bacteria							9	6	9	367	194	163						
Armatimonadetes bacterium JP3_11	Bacteria							1	1	2	54	20	29	2					
Thermus scotoductus	Bacteria							6	3	4	73	49	45	9	8	5			
Stenotrophomonas maltophilia	Bacteria										1							1	1
Lactobacillus mudanjiangensis	Bacteria																		
Hydrogenobacter hydrogenophilus	Bacteria	22	12	8	3	1	1	18	11	12	25	39	5	200	207	235			
Chloroflexus sp. Y-396-1	Bacteria							12	10	4	101	63	46		1	2			
Fischerella thermalis	Bacteria							3		2	57	30	27	2					
Chloroflexus sp. MS-G	Bacteria							12	2	10	55	43	36		1				
Aquificae bacterium	Bacteria	7	5	1			1	2	1		3	2		11	8	9			
Chloroflexi bacterium	Bacteria	1						3	4	2	59	36	19	4	1	8			
Sulfurihydrogenibium yellowstonense	Bacteria	37	11	14	9	4	6	8	4	7	12	21	12	69	63	72			
Sulfurihydrogenibium sp. YO3AOP1	Bacteria	7	2	1	4		3	11	3	6	3	12	2	4	6	16			
Candidatus Thermofonsia Clade 1 bacterium	Bacteria							4	4	8	1	2							
Thermodesulfobacterium geofontis	Bacteria	37	11	20	27	17	35	4	12	7	13	26	11	45	56	66			
Meiothermus cerbereus	Bacteria							6		5	8	12	7		1				
Synechococcus sp. 63AY4M1	Bacteria							2	2	3	82	30	40						
Hydrogenivirga sp. 128-5-R1-1	Bacteria	1	1	2	2	5	2	1	3		2	3	2	3	4	3			
Campylobacter coli	Bacteria																		
Caldilinea aerophila	Bacteria							6	5	4	47	33	21						
bacterium HR19	Bacteria	22	10	9	2	5	6	11	8	21	74	103	45	43	47	58			
Thermus sediminis	Bacteria				1			1	1	1	10		7		1				
Persephonella marina	Bacteria	2	4	1	1	1	3	2			1	1		10	4	5			
Clostridium sp. C105KSO15	Bacteria																		
Hydrogenobacter thermophilus	Bacteria	5	6	2	6	4	4	3	2	1	9	8	5	17	7	11			
Chlorobium sp. 445	Bacteria							7	5	5	15	16	7						
Thermocrinis albus	Bacteria	1	4	3	1	5	1	1	4		1	6		14	16	27			

Supplemental File ST3 (continued...)

Species	YNP61	YNP62	YNP65	YNP66	YNP69	YNP70	YNP73	YNP74	YNP77		YNP63	YNP64	YNP67	YNP68	YNP71	YNP72	YNP76	YNP78
	FSS3 (PC)		FSS1 (PC)		FSSe2 (PC)		Ose (PC)		PC control		FSS3 (Supor)		FSS1 (Supor)		FSSe2 (Supor)		OS (Supor)	Supor control
Pyrobaculum sp. WP30	85	480	290	334	449	105	32	58			577	169	313	325	358	677	461	
Pyrobaculum sp. OCT_11	5	49	25	20	25	3	2	4			54	9	28	21	33	45	41	
Thermoproteus sp. CP80	7	25	37	26	44	6	2				48	19	32	17	45	59	30	
Pyrobaculum sp. JCHS_4	3	29	15	15	35	5	2	3			46	7	24	16	26	52	27	
Pyrobaculum ferrireducens	3	23	15	12	22	2		2			21	5	11	11	21	35	21	
Thermoprotei archaeon	3	6	27	20	18	2	6	7			29	4	13	25	29	46	3	
Desulfurococcaceae archaeon AG1	5	6	8	9	6	1	1	4			16	4	10	9	20	40	5	
Pyrobaculum aerophilum	1	2	1	2	5	4					4	3	2	1	2	5	4	
Thermoproteus sp. JCHS_4	2	6	4	7	4	2					9	1	3	5	4	10	3	
archaeon HR02	3	7	22	19	29	4	11	5			9	7	14	12	17	12	6	
Caldivirga sp. MU80		5	6	7	13	1		1			10	2	7	7	2	11	1	
Candidatus Caldiarchaeum subterraneum		2	13	9	36	6	94	109	1		7	2	3	11	10	14	3	
Pyrobaculum islandicum	1	2	3	6	6	1					9	4	5	4	2	7	1	
Thermocrinis ruber	9844	71470	47491	47187	33795	4234	4982	5825			68397	14041	53314	54867	23731	35438	68063	5
Synechococcus sp. JA-2-3B'a(2-13)	1	1	35	75	5885	809	6944	9200			14		122	152	1029	2034	6	1
Thermus aquaticus	8	20	31	63	588	249	146	235			26	14	35	38	54	233	22	
bacterium HR17	312	276	2072	2422	1175	210	127	128	1		1417	347	502	593	225	395	338	
Clostridioides difficile								1					8	3				
Candidatus Thermochlorobacter aerophilum		1	1	91	188	44	1732	1986	1		59		32	31	3213	4799	3	2
Thermocrinis sp. GBS	94	684	503	506	362	48	48	65			682	147	567	583	283	417	665	
Armatimonadetes bacterium CP1_7O	3	2	22	30	234	83	156	266			14	8	8	12	70	208	1	
bacterium HR21	1	9	16	31	332	72	1194	1852			15	6	42	49	131	327	11	
Armatimonadetes bacterium DC	2	3	17	30	181	43	138	247			8	2	5	11	51	199	3	1
Synechococcus sp. JA-3-3Ab		2		3	426	74	365	510			5		5	8	107	295		
Bacteroidetes bacterium	5	1	17	23	456	74	497	617			13		20	20	441	757	2	9
Thermomicrobium roseum	1		3	3	190	44	243	462			4	1	3	14	12	39	1	
Armatimonadetes bacterium	5	4	6	29	149	55	91	149			5	2	4	6	40	109	6	1
Roseiflexus sp. RS-1			1	9	67	10	47	62			3		3	3	153	270		
bacterium HR16	5	5	37	42	165	32	374	517			34	5	17	20	140	261	5	
Synechococcus sp. 65AY6A5			1	2	332	52	308	406	1		1		3	10	113	205		
Thermoflexus hugenholtzii		2	4	3	15	2	22	34			5	1	4	5	5	6	1	
Gloeomargarita lithophora		1	6	57	195	34	92	122			45		2	4	2183	3190	1	5
Ralstonia solanacearum							2	2										
Lactobacillus rhamnosus																		
Synechococcus sp. 63AY4M2				1	324	44	286	370	1		2		2	8	79	163		
bacterium HR15	3	4	11	13	148	30	160	259			8	1	11	11	38	94	2	
Chloracidobacterium thermophilum		3	2	12	205	27	5	13			15		2	3	425	699	6	
Acidobacteria bacterium	3	15	28	19	232	35	503	534			20	4	13	18	271	360	10	61
Synechococcus sp. 65AY6Li				3	140	21	142	163						2	39	98		
Armatimonadetes bacterium JP3_11		1	7	8	56	17	41	66					4	2	16	57	2	
Thermus scotoductus		1	5	11	89	24	40	72			15	1	6	5	16	51	11	
Stenotrophomonas maltophilia									2									244
Lactobacillus mudanjiangensis																		
Hydrogenobacter hydrogenophilus	18	238	18	10	56	2	34	31			54	7	10	9	42	57	268	
Chloroflexus sp. Y-396-1				6	46	11	119	135			1		3	2	87	165		
Fischerella thermalis	1		4	4	33	10	37	55					2	5	17	45		
Chloroflexus sp. MS-G				3	52	15	101	108			1		1	2	87	157		
Aquificae bacterium	1	7	4	8	16	1	6	4			16	1	5	6	12	16	10	
Chloroflexi bacterium	3	6	8	9	20	5	37	59			7	1	2	6	10	40	9	11
Sulfurihydrogenibium yellowstonense	12	69	43	40	69	1	16	20			103	22	46	66	59	69	80	
Sulfurihydrogenibium sp. YO3AOP1	11	9	7	8	34	7	8	11			40	12	8	9	23	33	12	
Candidatus Thermofonsia Clade 1 bacterium					32	10	16	43			2				9	19		1
Thermodesulfobacterium geofontis	4	48	125	113	31	1	13	8			100	21	112	123	67	56	67	
Meiothermus cerbereus			10	9	35	11	15	27			4			1	22	54		
Synechococcus sp. 63AY4M1		1			38	3	28	33			2		1		14	33		
Hydrogenivirga sp. 128-5-R1-1	3	1	15	14	6		1	2			12	5	14	17	10	12	1	
Campylobacter coli																		
Caldilinea aerophila					29	8	21	26						1	12	28		
bacterium HR19	17	35	74	72	39		47	47			77	15	39	31	81	79	41	
Thermus sediminis		1	1	1	11	6	3	7			1			1	2	4	1	
Persephonella marina	5	7	19	18	11	1		1			17	2	29	24	10	14	2	
Clostridium sp. C105KSO15																		
Hydrogenobacter thermophilus	2	6	24	16	12	4	7	10			16	3	8	15	9	15	2	
Chlorobium sp. 445				1	11		62	70			3		2	1	110	147		
Thermocrinis albus	6	18	23	26	17		4	2			33	12	26	31	14	19	27	

Supplemental File ST4. Illumina-based shotgun metagenomic sequencing reads assigned to microbial taxa in YNP samples collected using different filtration methods.

Species	Domain	JY01	JY02	JY03	JY04	JY05	JY06	JY07	JY08	JY09	JY46	JY47	JY48	JY55	JY56	JY57	JY40	JY41	JY42
		FSS3 (Sterivex)			FSS1 (Sterivex)			FSSe2 (Sterivex)			OSe (Sterivex)			OS (Sterivex)			Sterivex control		
Pyrobaculum sp. WP30	Archaea	30532	28950	42968	48069	31903	38391	28880	22382	73942	1041	1110	691	25893	33787	30872	2	1	883
Pyrobaculum sp. OCT_11	Archaea	3357	3228	4877	4353	2787	3448	3200	2540	8665	120	144	91	2986	3943	3602			81
Thermoproteus sp. CP80	Archaea	3216	3047	4566	5040	3200	3884	3218	2514	8533	66	95	54	1748	2353	2236			86
Pyrobaculum sp. JCHS_4	Archaea	2513	2331	3561	4002	2579	3129	2457	1972	6246	78	75	44	1717	2255	2120			82
Pyrobaculum ferrireducens	Archaea	1616	1494	2338	2420	1621	1911	1550	1179	4116	61	54	36	1276	1733	1553			42
Thermoprotei archaeon	Archaea	1133	1028	1613	1528	1043	1193	1531	1234	4539	53	55	24	246	273	278			32
Desulfurococcaceae archaeon AG1	Archaea	966	845	1386	1095	669	797	1474	1127	4411	24	33	18	189	253	227			21
Pyrobaculum aerophilum	Archaea	665	635	899	1001	677	737	626	463	1529	24	17	14	503	654	631			20
Thermoproteus sp. JCHS_4	Archaea	533	511	723	854	555	677	491	416	1312	12	17	13	318	412	368			10
archaeon HR02	Archaea	347	328	515	533	353	413	484	366	1509	83	98	49	245	324	268	10		20
Caldivirga sp. MU80	Archaea	484	496	702	746	507	652	449	382	1325	8	10	11	313	352	352			11
Candidatus Caldiarchaeum subterraneum	Archaea	118	102	160	223	155	188	402	286	1164	649	735	408	79	95	100	6		36
Pyrobaculum islandicum	Archaea	430	379	553	648	417	569	401	344	1191	14	16	12	285	364	336			11
Thermoproteus sp. CIS_19	Archaea	391	353	566	654	421	485	354	252	929	17	5	5	245	308	262			11
Thermocrinis ruber	Bacteria	898309	813529	1271491	1514682	955539	1227329	309891	250761	874206	41418	42217	24908	1111169	1478378	1357995	38	21	25477
Synechococcus sp. JA-2-3B'a(2-13)	Bacteria	40	35	37	566	362	453	82958	59489	186125	434537	354555	221624	904	1222	1071	116	16	247
Thermus aquaticus	Bacteria	1683	1431	2286	5285	3284	3963	11066	8193	28028	18439	21440	15730	6134	7333	5943	36	2	93
bacterium HR17	Bacteria	4001	3673	5762	6975	4412	5458	3430	2537	8901	1092	1217	716	6809	10984	9624	14		116
Clostridioides difficile	Bacteria	3167	4612	22128	18465	4166	13230	6777	1881	11065	1414	2855	1470	14656	5090	3664	2633	452	10320
Candidatus Thermochlorobacter aerophilum	Bacteria	11	6	10	30	13	31	11731	9015	28599	6993	5885	3556	25	44	36	25	5	181
Thermocrinis sp. GBS	Bacteria	9407	8383	13507	15728	9916	12481	3833	3035	10491	516	525	277	10840	14369	13369			277
Armatimonadetes bacterium CP1_7O	Bacteria	334	309	464	1273	796	915	7197	4713	17281	25212	25733	14908	706	922	865	9	1	86
bacterium HR21	Bacteria	126	103	161	551	382	410	2754	1969	8965	14003	17227	9690	287	469	420	12		71
Armatimonadetes bacterium DC	Bacteria	196	166	271	722	459	512	4424	3017	10763	15206	15409	8675	394	586	534	11	1	54
Synechococcus sp. JA-3-3Ab	Bacteria	4		1	8	6	8	6481	4646	14755	27636	22748	14247	60	91	58	6	2	25
Bacteroidetes bacterium	Bacteria	72	68	96	721	453	484	10852	8276	26983	6297	5665	3515	74	81	51	13	9	101
Thermomicrobium roseum	Bacteria	49	56	88	93	50	73	2488	1824	6411	17630	19709	11811	414	593	559	27		87
Armatimonadetes bacterium	Bacteria	135	141	177	506	322	402	2947	1896	7082	10218	10191	5992	319	393	342	8	1	119
Roseiflexus sp. RS-1	Bacteria	15	13	23	54	22	15	3502	2473	8402	18884	15374	9148	180	263	223	51	18	2054
bacterium HR16	Bacteria	62	81	119	145	92	143	1170	839	3298	8535	8625	4983	115	170	149	18	1	141
Synechococcus sp. 65AY6A5	Bacteria	1	1	2	4	2	3	3199	2283	7162	13730	11209	6966	25	34	46	2		15
Thermoflexus hugenholtzii	Bacteria	518	499	787	2401	1573	1705	677	488	1614	5683	5044	3308	2128	2797	2414	28		42
Gloeomargarita lithophora	Bacteria	2		8	27	22	26	2044	1525	4840	1300	986	672	17	24	15	3	1	67
Ralstonia solanacearum	Bacteria	429	822	773	236	2622	1152	2009	1376	9097	2221	1754	620	3087	3736	3959	6	910	1372
Lactobacillus rhamnosus	Bacteria	2288	3036	157	315	983	594	1105	994	3707	4150	1743	1261	1733	2848	1253	5	82	406
Synechococcus sp. 63AY4M2	Bacteria	1	1	1	3	5	1	2846	1949	6356	12081	9812	6133	24	29	45	7		6
bacterium HR15	Bacteria	38	49	78	184	123	139	1667	1121	4218	5840	5852	3383	137	144	142	13		38
Chloracidobacterium thermophilum	Bacteria	9	11	18	27	27	21	2441	1731	5697	399	313	174	32	37	31	2		82
Acidobacteria bacterium	Bacteria	529	438	741	990	579	732	2896	2299	7176	1836	1575	905	446	614	531	83	88	614
Synechococcus sp. 65AY6Li	Bacteria				4	3	3	2109	1527	4914	9918	8055	5182	31	24	22	2	2	8
Armatimonadetes bacterium JP3_11	Bacteria	52	29	78	194	120	130	1107	804	2790	4134	4201	2523	117	142	129	1		19
Thermus scotoductus	Bacteria	181	177	259	596	356	420	1282	969	3323	2214	2558	1712	735	906	780	3		11
Stenotrophomonas maltophilia	Bacteria	111	80	154	113	63	83	73	45	169	60	69	50	313	340	352	3574	2958	14897
Lactobacillus mudanjiangensis	Bacteria	404	156	108	230	1390	160	298	101	4595	210	154	3262	406	174	1353	33	539	100
Hydrogenobacter hydrogenophilus	Bacteria	966	855	1419	1075	684	898	780	657	2327	106	93	75	1691	2266	1898			29
Chloroflexus sp. Y-396-1	Bacteria	5	6	7	29	17	17	1523	1157	3867	2073	1703	1079	18	18	19	5		62
Fischerella thermalis	Bacteria	19	25	37	103	61	79	842	587	2224	2670	2771	1607	67	77	75			12
Chloroflexus sp. MS-G	Bacteria	1	3	3	9	1	10	1441	1160	3665	2070	1747	1056	18	27	24			36
Aquificae bacterium	Bacteria	903	832	1329	1546	976	1280	540	409	1308	77	82	44	1095	1424	1272			30
Chloroflexi bacterium	Bacteria	58	67	87	178	122	158	913	680	2450	2981	2686	1641	79	104	96	27	20	236
Sulfurihydrogenibium yellowstonense	Bacteria	1039	936	1489	1323	847	1151	357	307	1069	53	56	53	595	758	651			30
Sulfurihydrogenibium sp. YO3AOP1	Bacteria	1172	1137	1665	907	554	701	961	827	2681	116	119	83	536	694	654			46
Candidatus Thermofonsia Clade 1 bacterium	Bacteria	2	1	9	5	2	5	1702	1177	3893	29	39	16	5	8	3			2
Thermodesulfobacterium geofontis	Bacteria	557	502	862	1187	616	959	186	152	538	27	30	10	348	372	287			24
Meiothermus cerbereus	Bacteria	40	37	64	191	112	155	673	462	1683	351	437	278	150	197	167	1	1	12
Synechococcus sp. 63AY4M1	Bacteria		1		4	2		739	531	1561	2972	2503	1518	5	7	9			3
Hydrogenivirga sp. 128-5-R1-1	Bacteria	539	478	777	1093	694	904	182	141	512	16	27	16	375	495	482			18
Campylobacter coli	Bacteria	40	55	19	2250	182	25	217	34	944	15	19	50	367	88	94	332	10	7
Caldilinea aerophila	Bacteria	15	7	6	27	17	27	853	672	2111	1476	1315	770	28	33	40	2		58
bacterium HR19	Bacteria	394	264	628	276	126	192	168	132	546	176	227	136	384	469	374	6		32
Thermus sediminis	Bacteria	39	57	71	165	86	122	455	350	1129	797	1035	698	233	282	235	1		6
Persephonella marina	Bacteria	429	427	603	695	468	515	166	118	419	27	33	12	412	568	555			14
Clostridium sp. C105KSO15	Bacteria	206	103	45	458	83	1909	78	219	391	130	83	38	123	477	123	1	6	1864
Hydrogenobacter thermophilus	Bacteria	378	358	565	867	569	737	317	288	881	72	63	45	286	403	318			9
Chlorobium sp. 445	Bacteria			1	2	2	1	464	329	1080	229	221	113	1	2	1	1		3
Thermocrinis albus	Bacteria	430	398	656	745	359	583	189	115	475	21	25	7	257	301	238			14
Thermus sp. YS45	Bacteria	40	36	54	146	102	102	332	285	1104	635	752	474	223	253	229	1		7
Thermus thermophilus	Bacteria	42	28	51	144	75	95	327	226	822	574	631	448	130	158	133			4
Candidatus Acetothermum autotrophicum	Bacteria	3	5	7	3	7	3	262	204	676	763	823	523	6	11	10	14		39
bacterium HR14	Bacteria	6	5	7	30	16	23	247	182	665	891	849	507	23	29	34			5
Candidatus Thermofonsia Clade 3 bacterium	Bacteria			1				296	225	835	1171	1094	669			1			1
Synechococcus sp. 65AY640	Bacteria							355	255	902	1646	1401	865	8	9	2	1		2

Supplemental File ST4 (continued...)

Species	JY61	JY62	JY65	JY66	JY69	JY70	JY73	JY74	JY77		JY63	JY64	JY67	JY68	JY71	JY72	JY76	JY78
	FSS3 (PC)		FSS1 (PC)		FSSe2 (PC)		Ose (PC)		PC		FSS3 (Supor)		FSS1 (Supor)		FSSe2 (Supor)		OSe (Supor)	Supor
Pyrobaculum sp. WP30	32685	15620	46545	38461	28206	23734	2298	1946	1		36573	48356	32014	29061	23471	22626	31918	9
Pyrobaculum sp. OCT_11	3430	1976	4246	3500	3085	2640	246	243	1		3876	4965	2987	2716	2622	2497	3614	2
Thermoproteus sp. CP80	3204	1164	5076	4092	3004	2501	166	151			3627	4871	3422	3066	2432	2419	2265	1
Pyrobaculum sp. JCHS_4	2549	1174	3833	3011	2185	1887	144	138			2820	3621	2638	2274	1902	1787	2240	
Pyrobaculum ferrireducens	1806	806	2346	2069	1611	1293	118	97			1945	2452	1631	1519	1174	1157	1628	
Thermoprotei archaeon	994	201	1146	1103	693	444	129	103	1		1031	1266	684	594	657	601	297	
Desulfurococcaceae archaeon AG1	739	184	682	618	549	248	55	56			794	965	382	329	476	569	244	
Pyrobaculum aerophilum	698	325	999	839	588	500	52	22	1		721	966	683	647	450	461	663	
Thermoproteus sp. JCHS_4	578	221	834	704	490	413	18	31			606	793	576	486	394	416	378	
archaeon HR02	596	164	1099	911	888	638	126	151			461	596	597	567	392	374	273	
Caldivirga sp. MU80	524	205	757	637	485	387	21	15	1		574	770	497	450	388	363	393	
Candidatus Caldiarchaeum subterraneum	167	54	419	382	861	647	1313	1291			148	178	255	196	278	296	100	
Pyrobaculum islandicum	435	186	648	494	357	313	26	33			431	616	430	347	261	325	353	
Thermoproteus sp. CIS_19	400	150	574	511	290	274	17	10			426	550	354	371	269	305	315	
Thermocrinis ruber	1717793	1300921	2203095	1817534	1001156	478701	92876	96228	178		1718244	2030445	1891898	1759308	571935	521878	1706989	265
Synechococcus sp. JA-2-3B'a(2-13)	295	286	5597	4783	200647	153202	290326	252881	12		110	1876	272	269	69840	60220	575	75
Thermus aquaticus	8098	2432	18797	15542	48044	39832	32954	25847	7		6249	9012	7838	7304	22228	19730	7164	26
bacterium HR17	37346	5085	59434	43834	23255	18478	2544	2556	9		22867	28310	10497	9669	6652	5938	10557	6
Clostridioides difficile	16024	4647	18749	22371	9349	3913	6842	6392	713		43183	38364	15238	6125	7375	2541	29350	617
Candidatus Thermochlorobacter aerophilum	14	22	68	43	9651	6557	41311	42466	2		16	261	30	38	89017	77926	43	77
Thermocrinis sp. GBS	17323	12383	22302	18588	10545	4501	1026	1121	3		17445	20202	19144	17901	6061	5605	16484	2
Armatimonadetes bacterium CP1_7O	2888	672	5509	4244	34701	24772	41809	34826	5		1912	2851	2126	1870	25387	22305	1791	46
bacterium HR21	687	316	2329	1914	10772	7842	46768	41402	1		634	1154	1241	1129	8804	7881	784	5
Armatimonadetes bacterium DC	1626	435	3256	2505	20419	14980	24870	21340			1087	1642	1182	1028	15318	13165	1108	40
Synechococcus sp. JA-3-3Ab	20	20	99	78	15836	11964	18835	16497			4	107	7	7	8956	7523	39	12
Bacteroidetes bacterium	148	48	964	819	12856	10000	12620	11846	3		95	204	418	347	12100	10907	73	22
Thermomicrobium roseum	488	178	547	424	10119	7710	22696	21937			267	513	195	183	2143	1923	432	4
Armatimonadetes bacterium	1170	273	2243	1748	13933	10126	16629	14254			780	1209	845	728	10022	8742	723	23
Roseiflexus sp. RS-1	27	77	60	45	4373	3279	2769	2521	5		24	84	18	18	9288	8267	139	16
bacterium HR16	690	81	827	622	6091	4673	14628	13769	2		376	622	190	162	6903	6365	187	17
Synechococcus sp. 65AY6A5	3	13	68	47	7699	5785	9359	8089			2	60	6	5	4210	3469	22	2
Thermoflexus hugenholtzii	3162	1036	6555	5262	2196	1876	10985	8560			1646	2402	3092	2775	1047	911	3052	
Gloeomargarita lithophora	37	7	511	393	4489	3671	867	832	5		14	27	39	51	29507	25080	13	58
Ralstonia solanacearum	2478	1047	9016	1922	8094	3813	1662	2107	1059		1484	743	3422	4256	2794	770	346	95
Lactobacillus rhamnosus	10356	1748	4740	1371	2914	8424	5235	2638	243		381	864	1872	3202	3500	1983	499	9
Synechococcus sp. 63AY4M2	5	4	51	41	6906	5102	8091	7044			3	45	4	9	3750	2974	18	3
bacterium HR15	489	126	838	695	9173	6921	10731	9788	1		307	475	336	287	4823	4349	341	3
Chloracidobacterium thermophilum	28	19	91	88	7568	5497	522	467			15	27	31	24	21389	18228	42	30
Acidobacteria bacterium	1024	508	1627	1361	5538	2611	6430	7095	32		1033	1177	1238	1133	4255	3548	710	100
Synechococcus sp. 65AY6Li	5	5	48	34	5109	3818	6476	5817			1	36	6	3	2448	2141	23	1
Armatimonadetes bacterium JP3_11	455	95	898	671	5625	4054	6785	5496	1		278	469	313	305	4262	3639	294	5
Thermus scotoductus	717	468	1663	1445	4862	3910	3929	3285	2		588	832	807	728	2009	1832	951	
Stenotrophomonas maltophilia	506	36	630	352	392	351	29	18	10793		565	747	270	226	83	75	86	5702
Lactobacillus mudanjiangensis	1727	1716	5137	303	3750	468	1224	2474	459		183	759	457	181	353	4333	399	
Hydrogenobacter hydrogenophilus	1667	2134	1616	1292	1367	463	201	232			1571	1807	1259	1247	768	673	2679	
Chloroflexus sp. Y-396-1	54	8	110	106	2777	2031	4858	4646	1		35	71	50	45	4048	3576	10	3
Fischerella thermalis	221	74	492	368	3702	2769	4424	3771	1		153	246	173	182	2703	2387	179	3
Chloroflexus sp. MS-G	14	17	31	18	2770	2106	4298	4079	1		8	32	7	7	3999	3661	18	9
Aquificae bacterium	1826	1284	2345	1809	1160	538	167	185			1794	1982	1789	1685	682	696	1694	
Chloroflexi bacterium	209	82	560	430	1565	1214	4613	3897	3		133	189	194	175	2442	2106	117	53
Sulfurihydrogenibium yellowstonense	2210	609	2020	1616	1427	468	148	210			2002	2339	1666	1597	686	654	978	
Sulfurihydrogenibium sp. YO3AOP1	2204	482	1120	1033	1848	790	242	289	1		1797	2043	883	833	960	877	667	
Candidatus Thermofonsia Clade 1 bacterium	9	4	39	31	3220	2191	5032	4191	1		3	40	11	13	2416	2029	4	1
Thermodesulfobacterium geofontis	804	457	1831	1572	639	115	40	60			1052	1105	1364	1496	426	353	603	1
Meiothermus cerbereus	163	71	1470	1155	1784	1320	1559	1242			87	144	241	218	2488	2181	251	4
Synechococcus sp. 63AY4M1	2	2	10	12	1680	1363	2089	1759			1	17	3	3	918	770	1	1
Hydrogenivirga sp. 128-5-R1-1	1203	297	1477	1193	597	324	53	64			1079	1362	1287	1210	365	318	610	
Campylobacter coli	311	160	875	59	1178	47	185	255	48		42	4617	443	93	27	77	3410	1
Caldilinea aerophila	41	11	90	67	1875	1460	1345	1169			18	48	33	39	1024	887	36	3
bacterium HR19	826	607	1150	858	819	112	303	457			946	1007	490	485	596	459	871	
Thermus sediminis	247	119	568	452	1684	1447	1436	1152			182	282	211	218	793	723	215	1
Persephonella marina	859	344	1057	891	527	360	44	44			877	998	1051	960	288	274	777	
Clostridium sp. C105KSO15	1172	210	379	2715	288	359	481	361	9		86	694	128	553	245	50	788	109
Hydrogenobacter thermophilus	690	223	991	853	612	261	116	124			587	739	610	534	337	337	325	
Chlorobium sp. 445			6		398	283	1489	1530				8	2		3485	3047		5
Thermocrinis albus	775	304	940	820	426	199	48	35			783	883	807	836	261	266	386	
Thermus sp. YS45	165	124	500	399	1335	1001	874	846			151	205	248	191	460	436	272	2
Thermus thermophilus	152	66	417	340	1228	973	985	870			149	174	196	185	568	510	176	
Candidatus Acetothermum autotrophicum	22	6	12	10	909	739	2833	2372			10	43	13	2	198	150	9	
bacterium HR14	65	20	112	77	1102	879	1623	1468			36	56	37	43	870	792	48	1
Candidatus Thermofonsia Clade 3 bacterium				1	509	406	1869	1502				16			1079	919		1
Synechococcus sp. 65AY640		3	7	2	887	741	1153	1028			1	9			551	503		